

Adam is one of the reasons why deep neural network can be trained relatively easily. However, it has a critical issues when implemented together of L_2 loss, a common regularization method.

Observation 1: incorporate L_2 loss in SGD is equivalent to weight decay

$$\mathcal{L}^*(\theta) = \underbrace{\mathcal{L}(\theta)}_{\text{loss with } L_2 \text{ regularization}} + \underbrace{\frac{\lambda}{2} \|\theta\|^2}_{\text{original loss}} \rightarrow L_2 \text{ regularization}$$

$$\Rightarrow \nabla \mathcal{L}^*(\theta) = \nabla \mathcal{L}(\theta) + \lambda \theta$$

$$\begin{aligned} \Rightarrow \theta^{(t+1)} &\leftarrow \theta^{(t)} - \eta \nabla \mathcal{L}^*(\theta^{(t)}) \\ &= \theta^{(t)} - \eta \nabla \mathcal{L}(\theta^{(t)}) - \lambda \theta^{(t)} \\ &= (1 - \lambda) \theta^{(t)} - \eta \nabla \mathcal{L}(\theta^{(t)}) \end{aligned}$$

Observation 2: L_2 loss in Adam does not result weight decay effect

Adam algorithm + L_2 loss

$$1. m^{(t+1)} \leftarrow \beta_1 m^{(t)} + (1 - \beta_1) \nabla \mathcal{L}^*(\theta^{(t)})$$

$$2. S^{(t+1)} \leftarrow \beta_2 S^{(t)} + (1 - \beta_2) \nabla \mathcal{L}^*(\theta^{(t)}) \otimes \nabla \mathcal{L}^*(\theta^{(t)})$$

$$3. \hat{m}^{(t+1)} \leftarrow \frac{m^{(t+1)}}{1 - \beta_1^t}$$

$$4. \hat{S}^{(t+1)} \leftarrow \frac{S^{(t+1)}}{1 - \beta_2^t}$$

$$5. \theta^{(t+1)} \leftarrow \theta^{(t)} + \alpha \frac{\hat{m}^{(t+1)}}{\sqrt{\hat{S}^{(t+1)} + \epsilon}}$$

it scales the gradients

effect: when apply L_2 regularization, we expect the gradients of the parameters with large values to be large. However, Adam scale down these gradients.

Solution: Adamw $\rightarrow$ Separates Adam and L_2 loss

the idea is very simple: instead of using $\nabla L^*(\theta)$ in step 1 and 2 we still use $\nabla L(\theta)$ (so L_2 regularization have no effect)

then we minus $\lambda\theta$ in step 5 (so we apply L_2 regularization directly to update parameters)

Adamw algorithm

$$1. m^{(t+1)} \leftarrow \beta_1 m^{(t)} + (1 - \beta_1) \nabla L(\theta^{(t)})$$

$$2. s^{(t+1)} \leftarrow \beta_2 s^{(t)} + (1 - \beta_2) \nabla L(\theta^{(t)}) \otimes \nabla L(\theta^{(t)})$$

$$3. \hat{m}^{(t+1)} \leftarrow \frac{m^{(t+1)}}{1 - \beta_1^t}$$

$$4. \hat{s}^{(t+1)} \leftarrow \frac{s^{(t+1)}}{1 - \beta_2^t}$$

$$5. \theta^{(t+1)} \leftarrow \theta^{(t)} + \alpha \hat{m}^{(t+1)} - \lambda \theta$$

It is then trivial to show Adamw have the weight decay effect.